Aviya Litman

AI4Science, computational genomics, deep learning, precision medicine, neurogenomics

Personal website | Email | GitHub | LinkedIn

Education

Princeton University *Princeton, NJ*
Ph.D. in Quantitative and Computational Biology *2022 - Present*
- Advisor: Professor Olga Troyanskaya
- Thesis: *AI-driven decomposition of genetic, phenotypic, and clinical heterogeneity*

Princeton University *Princeton, NJ*
Masters in Quantitative and Computational Biology *2022 - 2024*

Columbia University in the City of New York *New York, NY*
Bachelors in Mathematics *2018 - 2021*
- Cumulative GPA: 3.88

Experience

Troyanskaya Lab, Princeton University *Princeton, NJ*
PhD Candidate *September 2022 - Present*
- Developed a scalable framework to apply unsupervised mixture modeling for data-driven decomposition of phenotypic and clinical heterogeneity. Enabled the first robust, replicable classification of autism into distinct clinical subclasses with differing genetic and molecular underpinnings (published in *Nature Genetics*) [GitHub]
  - Work featured in *The Transmitter (Spectrum)* [press coverage]
- Developed **Otari**, the first sequence-based graph deep learning model for predicting transcriptome-wide, isoform-level regulatory, expression, and variant effect changes across 30 human tissues and brain regions. Trained on genomic sequence and long-read transcriptomic data, Otari revealed widespread variant-driven isoform dysregulation across large regulatory disease mutation datasets, including an autism cohort (under review) [GitHub]
- Leveraged sequence-based deep learning to characterize RNA binding protein dysregulation in a large chronic kidney disease cohort, highlighting its potential role in disease pathogenesis (abstract published in the *Journal of the American Society for Nephrology*)
- Advanced and engineered core features of **Selene**, a Python library and command line interface for training deep neural networks from biological sequence data [GitHub]

Flatiron Institute, Simons Foundation *New York, NY*
Summer Research Associate *Summers 2024 and 2025*
- Led key research efforts with the Center for Computational Biology and the Simons Foundation Autism Research Initiative (SFARI), advancing deep learning and statistical genomics approaches for studying autism. Work laid the foundation for ongoing doctoral projects.

Korem Lab, Columbia University Irving Medical Center *New York, NY*
DSI Research Scholar / Research Associate *October 2020 - August 2022*
- Advisor: Prof. Tal Korem
- Awarded competitive research funding by the Columbia Data Science Institute and selected as a Data Science Institute Scholar.
- Evaluated and applied **CoPTR**, a bioinformatics tool for analyzing temporal growth dynamics in the microbiome. Demonstrated its accuracy in predicting growth in natural bacterial populations, outperforming prior state-of-the-art methods (published in *Genome Research*).
- Developed **Copangraph**, a data structure, analytic framework, and suite of algorithms for modeling extensive genomic variability across diverse metagenomes (manuscript in preparation).

Publications

1. **Aviya Litman**, Natalie Sauerwald, LeeAnne Green Snyder, Jennifer Foss-Feig, Christopher Y. Park, Yun Hao, Ilan Dinstein, Chandra L. Theesfeld, and Olga G. Troyanskaya. *Decomposition of phenotypic heterogeneity in autism reveals underlying genetic programs*. *Nature Genetics* (2025). [manuscript]
2. **Aviya Litman**, Zhicheng Pan, Ksenia Sokolova, Joyce Fang, Tess Marvin, Natalie Sauerwald, Christopher Y. Park, Chandra L. Theesfeld, and Olga G. Troyanskaya. *Variant-resolved prediction of context-specific isoform variation with a graph-based attention model*. Under review (2025).
3. Tyler A. Joseph, Philippe Chlenski, **Aviya Litman**, Tal Korem, and Itsik Pe'er. *Accurate and robust inference*

of microbial growth dynamics from metagenomic sequencing reveals personalized growth rates. Genome Research (2022) 32(3):558-568. [link]

4. Tess Marvin, **Aviya Litman**, Chen Wang, Chandra Theesfeld, Laura Mariani, Krzysztof Kiryluk, Matthias Kretzler, Olga Troyanskaya, and CureGN Genetics. *Characterizing the Genetic Architecture of Rare Glomerulonephropathy Disease: The Landscape of RNA Binding Protein Dysregulation. Journal of the American Society of Nephrology* (2023) 34(11S):p 416. [link]

Selected Talks

1. Princeton Precision Health Lecture Series, invited speaker. Princeton, NJ, March 2025
2. Winter Q-Bio Conference, main talk. Honolulu, Hawaii, February 2025
3. Lewis-Sigler Institute Symposium, main talk. Trenton, NJ, October 2024
4. Computational Systems for Integrative Genomics Conference, poster presentation. New York, NY, July 2024
5. NHGRI Research Training and Career Development Annual Meeting, poster presentation. Seattle, Washington, April 2024
6. American Society of Nephrology Conference, poster presentation. Philadelphia, Pennsylvania, November 2023

Awards

2024	**NSF GRFP (HM)**
2022	**Lewis-Sigler Institute Scholars Award**, Princeton University
2020-21	**Columbia Data Science Institute Scholar**, Research Fellowship ($10k)
2019	**Columbia Emerging Scholar in CS**
2019	**Columbia Engineering Fast Pitch Competition Finalist**
2018-21	**Dean's List for Academic Excellence**, all semesters

Outreach

Computational Systems for Integrative Genomics Conference — *Flatiron Institute, New York, NY*

CONFERENCE ORGANIZER — *July 2024*

- Co-led organization of a multi-day research conference including speaker recruitment, abstract review, scheduling, activities, and participant outreach for over 80 attendees from **21 domestic and international institutions**.

REU in Computational Biology for Previously Incarcerated Students — *Princeton, NJ*

VOLUNTEER INSTRUCTOR — *June 2023*

- Prepared and taught lesson on Python programming, data analysis, and ML as part of the REU program for previously incarcerated individuals at Princeton University [GitHub].

Columbia Undergraduate Admissions / Visitors Center — *New York, NY*

OUTREACH INTERN — *May 2019 - August 2019*

- Prepared and delivered presentations and STEM-themed campus tours to prospective students and their families.

Skills

Programming	Python, C/C++, Cython, MySQL, HTML/CSS, LaTeX, Bash
Data science	Numpy, Scipy, Pandas, Scikit-learn
Machine learning	PyTorch, Lightning, wandb
General	Linux/UNIX, Git, Vim, Slurm / Cluster environment
Languages	English, Hebrew